

Adjoint-based maximisation of heat flux in models of stellar convection

Applicant: Dr Laura Currie. **Undergraduate student:** Callum Wright-Parish

This is an application for a bursary to enable Callum Wright-Parish, an undergraduate student at Durham University, to undertake a 6-week summer research project from 29th June to 7th August 2026. The project will be supervised by the applicant, Dr Laura Currie, Assistant Professor in the Department of Mathematical Sciences.

Motivation and scientific background

Convective heat transport plays a central role in stellar evolution, governing the structure, luminosity, and lifetime of stars. In one-dimensional stellar evolution models, convection cannot be resolved explicitly and is instead usually parametrised using mixing length theory (MLT) [1]. Despite its widespread success, MLT rests on strong assumptions about the nature of turbulent convection, most notably that heat transport is dominated by a single, representative convective mode. Classical MLT estimates the convective heat flux by balancing the linear growth rate of buoyantly unstable motions against a phenomenological nonlinear cascade rate. In extensions of MLT that include rotation and/or magnetism (see e.g., [2]), this balance is used to select the mode that maximises the heat flux, under the assumption that the most efficient heat-transporting mode dominates the dynamics. Understanding which flow structures are most efficient at transporting heat is therefore a fundamental problem in stellar astrophysics.

Recent work by Sondak et al. [3] and others has begun to explore optimal heat transport solutions for classical Rayleigh–Bénard convection using variational methods, identifying flow structures that maximise the heat transport (as quantified by the Nusselt number). However, fundamental questions remain: how do optimal heat-transporting modes change when physical effects relevant to stellar interiors—rotation, magnetic fields, and stratification—are included? Do rotational and magnetic MLT extensions correctly identify modes that maximise heat transport under these constraints? We aim to address these questions in this project.

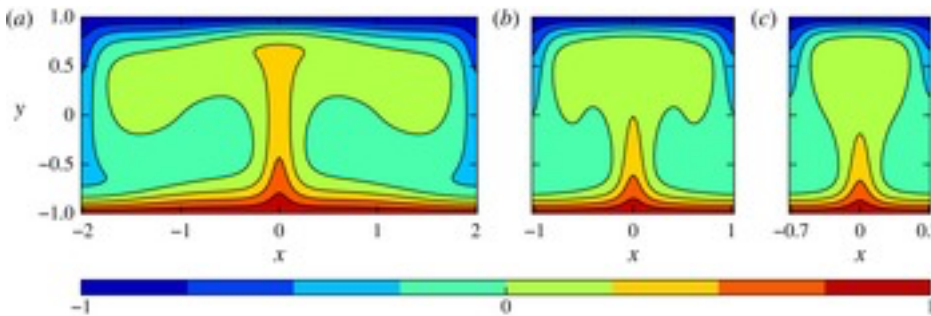


Figure 1. Temperature contours of different solutions in 2D Rayleigh–Bénard convection from [3]. In this case the solution in (c) is the optimal solution. In this project we will examine how optimal modes change in the presence of physical effects such as rotation, magnetic field and stratification.

Methodology

We will make use of adjoint-based optimisation methods applied to simulations of two-dimensional convection in a fluid layer. Despite its simplicity, this system captures essential features of stellar convection and has been extensively studied, which provides a well-controlled framework for studying convective heat transport.

Adjoint-based optimisation has become a powerful tool in fluid dynamics, allowing the systematic identification of optimal perturbations, flow structures, or forcing that extremise a given quantity (see e.g., [4]). Here we identify which velocity and temperature fields, subject to the governing equations of thermal convection, maximise the Nusselt number.

We will use *Dedalus* [5], an open-source Python-based code widely used for studying fluid dynamics and astrophysical flows. *Dedalus* allows users to specify the governing equations in a natural, symbolic form and then solves them numerically with high accuracy. Recent developments in the *Dedalus* ecosystem enable the systematic construction of adjoint equations corresponding to a given forward problem, allowing gradients of an objective functional (such as the Nusselt number) to be computed efficiently.

Callum will start by working through Dedalus tutorials to gain a basic understanding of the framework. He will then implement the adjoint framework for classical Rayleigh–Bénard convection in Dedalus and compare the results with optimal solutions from Sondak et al. [3] and Wen et al. [6] obtained via different methods.

This comparison alone could prove interesting; if differences occur, we will explore why. On the other hand, if the results here agree with those obtained via different methods, we will go on to explore how optimal modes change when stellar-relevant physics is introduced. Depending on progress and student interests, Callum may investigate how (i) rotation, (ii) magnetic fields, or (iii) stratification modify flow patterns that maximize heat transport. For each extension, optimal modes will be compared with those assumed in corresponding MLT formulations [2]. While optimisation studies exist for classical convection, systematic exploration of how rotation, magnetism, and stratification modify optimal structures has not been undertaken.

Feasibility

This project is well suited to an undergraduate researcher because the numerical framework is already developed and well-documented, with easy-to-follow tutorials. Callum will primarily interact with Dedalus through Python scripts, building on existing examples and documentation. The 2D system is computationally tractable on standard workstations, and the problem is modular with achievable initial goals and natural extensions should good progress be made. Dedalus's flexibility means adding rotation, magnetic fields, and/or stratification requires only modest modifications to governing equations rather than fundamental algorithmic changes.

Research environment

The project will furnish Callum with hands-on experience of modern, research-grade computational tools that are actively used in stellar astrophysics, while developing an understanding in fluid dynamics, numerical optimisation techniques, and astrophysical applications.

Callum will be allocated a desk and computer in the Department of Mathematical Sciences, and the applicant will closely supervise his work. Throughout this project Callum will be encouraged to read existing literature and develop his understanding of convection theory and stellar astrophysics.

During the project, Callum will join the weekly seminar meetings of the Applied Mathematics research group. At the end of the project, Callum will write a short report documenting his findings and present his work to the Applied Mathematics research group in the Department of Mathematical Sciences. If sufficient progress is made, the work may lead to the submission of a journal article to *Monthly Notices of the Royal Astronomical Society* on which Callum would be a co-author.

Student statement

My academic journey is defined by a deep passion for space exploration and an ambition to make meaningful contributions to this scientific frontier. I have always been captivated by the observation of stellar bodies. My experience in the university's Spaceflight Society has shifted my focus toward mathematical modelling. As we design a liquid-fuelled rocket for an upcoming launch, I have found myself thoroughly engaged by the application of theoretical mathematics to physical problems - particularly when calculating dynamic stability using matrix eigenvalues. I am eager to translate this enthusiasm for mathematical modelling to the study of stellar convection. The opportunity to explore the connection between rigorous theory and complex physical systems through adjoint-based maximisation is one I find truly exhilarating. I am confident that this project will not only provide valuable insight into applied mathematics research but also allow me to develop the rigorous research skills necessary for a career in astrophysics.

References

- [1] Böhm-Vitense, E., *ZA* **46**, 108, 1958.
- [2] Stevenson, D. J., *GApFD* **12**, 139, 1979.
- [3] Sondak, D., Smith, L.M., & Waleffe, F., *JFM* **784**, 565, 2015.
- [4] Kerswell, R.R., Pringle, C.C. & Willis, A.P., *Rep. Prog. Phys.* **77**, 085901, 2014.
- [5] Burns, K. J., Vasil, G. M., Oishi, J. S., Lecoanet, D., & Brown, B. P., *PhRvR* **2**, 023068, 2020.
- [6] Wen, B., Ding, Z., Chini, G.P., & Kerswell, R.R., *Philos. Trans. R. Soc. A* **380**, 2225, 2022.